

TestHub

Presentation

Assignment Summary

- **Overview of TestHub:** A WPF-based application for creating, managing, and analyzing tests.
- **Key Use Case:** Designed for educational and professional assessments with dynamic visualizations.
- **Unique Value:** Combines test management, user interaction, and data insights in a single application.

Implementation Remarks

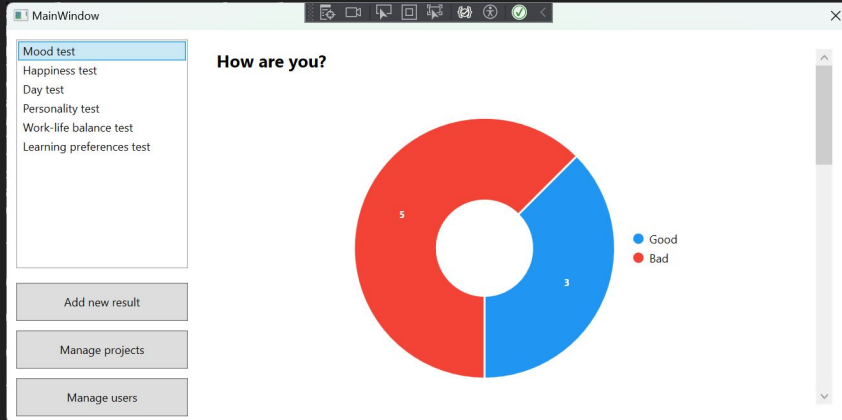
Functional Requirements:

- Test creation, management, and analysis.
- User management with real-time validation.
- Data visualization using dynamic charts.

Non-functional Requirements:

- Clean architecture using MVVM.
- Responsive and intuitive user interface.
- Performance optimization for database queries.

Screenshots and Visuals



Main window

The SelectUser window displays a list of users on the left and their details on the right. The list includes: timi, anni (selected), johnsmith, kate123, mike77, sara01, alex99, linda45, robertx, emma22, chris007, olivia56, daniel89, and amelia77. The details on the right show: Login: anni, Name: Anna, Surname: Ivanova, and E-mail: anna@gmail.com. At the bottom, there are buttons for Edit, OK, and Cancel.

Select user window

The Mood test window displays a series of questions and answers. The questions are: "How are you?", "Is it your best day today?", and "Are you an optimist or pessimist?". The answers are: "Good", "Yes", and ":D". At the bottom, there are buttons for OK and Cancel.

Test window

ManageProjects

Mood test

- Happiness test
- Day test
- Personality test
- Work-life balance test
- Learning preferences test

How are you?

Good
Bad

Is it your best day today?

Yes
No
Of course

Are you an optimist or pessimist?

:D
:(

Project title:
Mood test
Update

Question:
Add
Edit
Remove

Create Remove OK Cancel

Manage projects window

Manage users window

ManageUsers

timi
anni
johnsmith
kate123
mike77
sara01
alex99
linda45
robertx
emma22
chris007
olivia56
daniel89
amelia77

Login: : johnsmith

Name: John

Surname: Smith

E-mail: john.smith@gmail.com

Create Remove OK Cancel

Interesting Technical Solutions

Highlights:

- **MVVM Architecture:** Ensures scalability and maintainability.
- **Custom Control for Charts:** Pie charts dynamically update based on user interactions.
- **User Validation:** Prevent duplicate user logins with real-time checks.
- **SQLite Integration:** Efficient handling of test data and results.
- **ScrollView Optimization:** Smooth user experience with adjusted scroll speed.

Key Question: How does this project stand out?

- Integrates test creation, visualization, and user interaction seamlessly.
- Custom animations and styles enhance usability.

Summary of Project Impact:

- Provides an end-to-end solution for testing and analysis.
- Modular and reusable design for scalability.